

IMPACT OF WEATHER VARIABLES ON ELECTRICITY DEMAND: A COMPARATIVE MACHINE LEARNING STUDY



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Abstract

Accurate electricity demand forecasting is critical for the reliable and efficient operation of power grids, particularly in compact, highly urbanised nations such as Singapore where climate conditions remain consistently tropical and demand fluctuations are strongly coupled to weather patterns. Despite the growing availability of high-resolution meteorological data and advances in machine learning, systematic comparative studies that rigorously evaluate the predictive contribution of climatic variables to short-term electricity demand remain limited for this geographical context.

This study presents a comprehensive machine learning pipeline for daily electricity demand forecasting in Singapore, leveraging weather-derived and temporal features extracted from data spanning February 2023 to December 2025. A feature set of 16 variables was constructed, encompassing temperature, humidity, a Comfort Climate Index (CCI), calendar-based indicators (day of week, month, season, public holidays, and weekends), and historical demand lag and rolling statistics. Five forecasting models were trained and evaluated under a chronological 80/20 train-test split: Linear Regression, Random Forest, XGBoost, CatBoost, and a novel CatBoost model optimised with Phasor Particle Swarm Optimisation (PPSO). Models were assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), Mean Absolute Percentage Error (MAPE), Relative Absolute Error (RAE), and Willmott Index (WI).

The PPSO-optimised CatBoost model achieved the strongest predictive performance, attaining an R^2 of 0.911, an MAE of 1,520.2 MWh, and an RMSE of 1,922.1 MWh on the held-out test set, outperforming all baseline models. These results demonstrate that automated hyperparameter tuning via PPSO yields meaningful gains in gradient-boosted tree models and that weather-derived features, particularly temperature and its interaction with humidity, are significant predictors of electricity demand in Singapore. The findings have practical implications for grid operators and energy planners seeking data-driven approaches to short-term load forecasting.

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Declaration

I confirm that the work contained in this Degree project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks. **The declaration can be found inside template.cls. Make sure you put your name and adjust the date in the dotted lines!**

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Chapter 1

Introduction

Electricity is a fundamental driver of modern economic activity and daily life. In densely urbanised nations such as Singapore, where the tropical climate sustains year-round cooling demand and a compact grid serves approximately 5.9 million residents, accurate short-term forecasting of electricity demand is essential for reliable grid operation, cost-effective generation scheduling, and long-term infrastructure planning (Energy Market Authority of Singapore 2025). This chapter introduces the research context, states the problem addressed by this project, defines the objectives, and outlines the structure of the remaining chapters.

1.1 Background and Motivation

Global energy consumption has risen steadily over the past decades, driven by population growth, urbanisation, industrial expansion, and the proliferation of electrical appliances (Ringkjøb, Haugan, and Solbrekke 2018). Singapore, as a city-state with no indigenous fossil-fuel resources, imports virtually all of its primary energy and converts a substantial share into electricity. The Energy Market Authority (EMA) reported that national electricity consumption exceeded 55 TWh in 2024, with the residential and commercial sectors collectively accounting for the majority of the load (Energy Market Authority of Singapore 2025).

Unlike temperate regions where heating demand dominates winter consumption, Singapore’s equatorial climate produces a near-constant ambient temperature of 25–32°C and relative humidity of 70–90%. Air-conditioning therefore represents the single largest end-use of electricity in buildings, making demand highly sensitive to even small variations in temperature and humidity (Zhao and Magoulès 2012). Public holidays, week-ends, and monsoon seasons introduce additional variability that compounds the weather

effect.

Accurate demand forecasting allows grid operators to schedule generation units efficiently, minimise spinning reserves, negotiate fuel contracts, and defer capital-intensive capacity additions. Conversely, persistent forecast errors lead to over-generation losses, increased reliance on expensive peaking plants, or, in extreme cases, load shedding. The economic and operational consequences underscore the need for forecasting models that capture the complex, non-linear interactions between meteorological conditions and electricity demand.

1.2 Problem Statement

Traditional demand-forecasting approaches—including exponential smoothing, ARIMA, and simple regression—assume linear or stationary relationships between predictors and load. These methods perform adequately when consumption patterns are stable and the number of explanatory variables is small. However, Singapore’s electricity demand is shaped by a combination of weather variables (temperature, humidity, solar radiation, wind speed), temporal patterns (day-of-week, month, season, public holidays), and autoregressive dynamics (recent demand levels). The interactions among these factors are inherently non-linear and time-varying, rendering classical statistical methods insufficient for high-accuracy short-term forecasting (Al-Musaylh et al. 2018).

Machine learning models—such as Random Forest, XGBoost, and CatBoost—are well suited to capturing these non-linear relationships but introduce the challenge of hyperparameter tuning. The predictive performance of gradient-boosted tree models is known to be sensitive to choices such as tree depth, learning rate, number of iterations, and regularisation strength. Manual or grid-search tuning is computationally expensive and may converge to sub-optimal configurations.

Zhang et al. (Zhang, Y. Chen, and Yan 2023) demonstrated that combining CatBoost with a Phasor Particle Swarm Optimisation (PPSO) algorithm yields superior forecasting accuracy compared to CatBoost paired with five other meta-heuristic optimisers. However, their study was conducted on hourly residential data from a single customer in a different climate context. The extent to which this hybrid approach generalises to daily national-level demand in a tropical setting remains unexplored.

1.3 Research Objectives

This project addresses the identified gaps by pursuing the following objectives:

1. **Data Integration:** Construct a comprehensive daily dataset for Singapore by merging electricity demand records from EMA with weather observations from the Open-Meteo historical weather API, spanning February 2023 to December 2025.
2. **Feature Engineering:** Design a feature set that captures weather conditions, derived comfort indices, temporal patterns, and autoregressive demand dynamics.
3. **Comparative Modelling:** Train and evaluate five forecasting models—Linear Regression, Random Forest, XGBoost, CatBoost, and CatBoost-PPSO—under a consistent chronological train/test split.
4. **Hyperparameter Optimisation:** Implement the PPSO meta-heuristic to automatically tune the hyperparameters of CatBoost and assess the performance gain relative to default and standard CatBoost.
5. **Comprehensive Evaluation:** Compare all models using six statistical metrics (MAE, RMSE, R^2 , MAPE, RAE, and Willmott Index) and analyse feature importance to identify the most influential predictors of electricity demand.
6. **Interactive Dashboard:** Develop a web-based dashboard that visualises the historical demand data and exposes the best-performing trained model through a simple date-based prediction interface for non-technical users.

1.4 Scope and Limitations

The scope of this project is limited to *daily* electricity demand forecasting at the *national* level for Singapore. The study does not address intra-day (hourly or sub-hourly) forecasting, sector-specific demand decomposition, or real-time operational deployment. Weather data is obtained retrospectively from the Open-Meteo archive rather than from a live forecasting service; consequently, the pipeline evaluates model accuracy on historical data rather than genuine out-of-sample future predictions.

The CatBoost-PPSO implementation in this project follows the methodology proposed by Zhang et al. (Zhang, Y. Chen, and Yan 2023) with adaptations for daily resolution and the Singapore context. Deep learning approaches (e.g. LSTM, Transformer) are outside the scope of this comparative study, although they are discussed in the literature review for completeness.

1.5 Chapter Overview

The remainder of this report is organised as follows:

Chapter 2 *Background Research*. Reviews the literature on electricity demand forecasting, the CatBoost algorithm, and phasor particle swarm optimisation.

Chapter 3 *Design*. Presents the overall system architecture, data acquisition strategy, feature engineering design, and the rationale for model selection.

Chapter 4 *Implementation*. Describes the technical implementation of the data pipeline, preprocessing modules, model training, and PPSO-based hyperparameter tuning.

Chapter 5 *Evaluation & Testing*. Reports experimental results, model comparison, feature importance analysis, and discusses the findings.

Chapter 6 *Conclusion*. Summarises the key contributions, identifies limitations, and suggests directions for future research.

1.6 Conclusion

This chapter has established the motivation for weather-driven electricity demand forecasting in Singapore, articulated the research problem, and defined five concrete objectives. The following chapter examines the theoretical and empirical background underlying the methods employed in this project.

Chapter 2

Background Research

This chapter provides the theoretical foundation and literature context for the forecasting methods used in this project. It begins with an overview of electricity demand forecasting, examines the influence of weather variables, reviews classical and machine-learning approaches, and then presents in-depth descriptions of the two core techniques—CatBoost and Phasor Particle Swarm Optimisation (PPSO)—that form the proposed hybrid model.

2.1 Electricity Demand Forecasting: An Overview

Electricity demand forecasting is the process of estimating future electricity consumption based on historical data, weather conditions, and other contextual variables. Forecasts are categorised by their prediction horizon (Amasyali and El-Gohary [2018](#)):

- **Very short-term** (minutes to a few hours): used for real-time grid balancing.
- **Short-term** (one day to one week): used for generation scheduling and fuel procurement.
- **Medium-term** (one month to one year): used for maintenance planning and contract negotiation.
- **Long-term** (one year and beyond): used for infrastructure investment and policy formulation.

This project focuses on *short-term daily* forecasting, where the objective is to predict the total electricity demand (in MWh) for the next day. Short-term forecasts are critical because they directly influence operational decisions such as unit commitment, spinning reserve allocation, and spot-market trading strategies (Akay and Atak [2007](#)).

2.2 Role of Weather Variables in Electricity Demand

Weather conditions are among the most significant external drivers of electricity consumption, particularly in climate-sensitive regions. The key weather variables and their effects on demand are summarised below.

2.2.1 Temperature

Temperature has the most direct and well-documented impact on electricity consumption. In tropical climates such as Singapore, higher ambient temperatures increase air-conditioning load, which constitutes a substantial portion of total electricity demand. The relationship between temperature and demand is non-linear: moderate temperature variations produce limited changes in consumption, whereas extreme heat triggers sharp demand spikes as cooling systems operate at full capacity (Zhao and Magoulès 2012). In this project, the daily mean air temperature at 2m height is used as the primary weather predictor.

2.2.2 Humidity

Relative humidity amplifies the perceived thermal discomfort beyond what temperature alone indicates. High humidity reduces the effectiveness of evaporative cooling, prompting occupants to increase air-conditioning usage even when air temperatures are moderate. The interaction between temperature and humidity is captured in this project through a *Comfort Climate Index* (CCI), defined as:

$$CCI = T + 0.55 \times (1 - RH/100) \times (T - 14.5) \quad (2.1)$$

where T is the mean daily temperature ($^{\circ}\text{C}$) and RH is the mean daily relative humidity (%). The CCI provides a single metric that approximates the combined thermal stress experienced by building occupants.

2.2.3 Solar Radiation and Wind Speed

Daily cumulative shortwave radiation (W/m^2) affects demand through its correlation with daytime temperatures and, in buildings with significant glazing, through direct solar heat gain. Wind speed influences the rate of convective heat exchange and may reduce or increase cooling demand depending on the prevailing conditions. Both variables are included in the feature set as secondary weather predictors.

2.2.4 The Singapore Context

Singapore’s equatorial location (1.35°N, 103.82°E) produces a consistently warm and humid climate with minimal seasonal temperature variation. However, the monsoon seasons—Northeast Monsoon (December–March) and Southwest Monsoon (June–September)—introduce variability in rainfall, cloud cover, and humidity patterns that can influence daily demand. Unlike temperate regions where heating load creates a U-shaped temperature–demand relationship, Singapore’s demand is monotonically increasing with temperature, simplifying the modelling task but also reducing the signal-to-noise ratio for marginal temperature changes.

2.3 Classical Forecasting Methods

2.3.1 ARIMA and Exponential Smoothing

Autoregressive Integrated Moving Average (ARIMA) models capture linear temporal dependencies by modelling the relationship between past observations and future values. These models require the time series to be stationary (or transformed to achieve stationarity) and are effective when demand follows regular, repetitive patterns. Exponential smoothing methods (e.g. Holt-Winters) assign greater weight to recent observations, allowing adaptation to trends and seasonality (Al-Musaylh et al. 2018).

However, both approaches assume linear relationships and cannot easily incorporate exogenous variables such as weather without extensions (e.g. SARIMAX). Their inability to model non-linear interactions limits their accuracy in complex multi-variable forecasting scenarios.

2.3.2 Regression-Based Methods

Linear and polynomial regression models extend classical forecasting by incorporating external predictors. These methods are interpretable and computationally inexpensive but struggle with the non-linear, threshold-dependent relationships between weather variables and electricity demand. As Schirmer et al. (Schirmer, Mporas, and Potamitis 2019) demonstrated, linear regression is consistently outperformed by tree-based methods in residential energy consumption prediction tasks.

2.4 Machine Learning for Demand Forecasting

Machine learning methods learn patterns directly from data without requiring explicit assumptions about functional form. The following subsections describe the specific algorithms used in this project.

2.4.1 Random Forest

Random Forest (Breiman 2001) is an ensemble method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees. Each tree is trained on a bootstrapped sample of the data, and at each split, a random subset of features is considered. This dual randomisation reduces variance and overfitting compared to a single decision tree.

Random Forest naturally captures non-linear relationships and feature interactions, handles mixed data types without scaling, and provides built-in measures of feature importance through permutation or impurity-based metrics. Its main limitation is that, as a bagging ensemble, it does not sequentially correct errors, which can result in higher bias compared to boosting methods.

2.4.2 XGBoost

Extreme Gradient Boosting (XGBoost) (T. Chen and Guestrin 2016) is a scalable implementation of gradient boosting that builds trees sequentially, with each new tree fitted to the residual errors of the previous ensemble. XGBoost incorporates regularisation terms (L_1 and L_2) in the objective function to control model complexity:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (2.2)$$

where l is the loss function, $\hat{y}_i$ is the prediction for observation i , f_k is the k -th tree, and $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ penalises tree complexity through the number of leaves T and leaf weight magnitude w .

XGBoost offers several computational advantages, including parallel tree construction, cache-aware access patterns, and built-in handling of missing values. It has achieved state-of-the-art results in numerous tabular prediction tasks and is widely used in energy forecasting (Khan, Byun, et al. 2020).

2.4.3 CatBoost

CatBoost (Categorical Boosting) (Prokhorenkova et al. 2018) is a gradient boosting framework developed by Yandex that introduces two key innovations:

1. **Ordered Boosting:** To address the prediction shift (also known as target leakage) that occurs in standard gradient boosting when the same data is used for both residual computation and tree construction, CatBoost uses a permutation-driven ordered boosting scheme. For each training example, the residual is computed

using a model trained only on preceding examples in a random permutation, thereby eliminating the bias.

2. **Native Categorical Feature Handling:** CatBoost encodes categorical features using *Target Statistics* (TS), where the encoding for each category is computed from the target values of preceding observations in the ordered permutation. This approach avoids the information leakage inherent in standard target encoding while preserving the predictive value of categorical variables.

The CatBoost prediction for an observation $\mathbf{x}$ is:

$$\hat{y}(\mathbf{x}) = \sum_{t=1}^T \eta \cdot h_t(\mathbf{x}) \quad (2.3)$$

where h_t is the t -th symmetric (oblivious) decision tree, η is the learning rate, and T is the total number of trees. Oblivious trees use the same splitting condition at each level, resulting in a balanced tree structure that is faster to evaluate and less prone to overfitting (Dorogush, Ershov, and Gulin 2018).

CatBoost is particularly well-suited to this project because the feature set includes several categorical variables—day of week, month, season, weekend indicator, and public holiday indicator—that benefit from native categorical handling rather than one-hot encoding.

Key Hyperparameters

The predictive performance of CatBoost is sensitive to four main hyperparameters:

Table 2.1: Key CatBoost hyperparameters and their roles.

Parameter	Range	Effect
<code>iterations</code>	100–800	Number of boosting rounds (trees)
<code>learning_rate</code>	0.01–0.20	Step size for gradient descent
<code>depth</code>	4–10	Maximum depth of each oblivious tree
<code>l2_leaf_reg</code>	1.0–20.0	L_2 regularisation on leaf values

Selecting optimal values for these parameters is critical: too many iterations or excessive depth can lead to overfitting, while too few iterations or aggressive regularisation may underfit the data. This motivates the use of an automated meta-heuristic optimiser, as described in the following section.

2.5 Phasor Particle Swarm Optimisation (PPSO)

2.5.1 Standard Particle Swarm Optimisation

Particle Swarm Optimisation (PSO) (Kennedy and Eberhart 1995) is a population-based meta-heuristic inspired by the social behaviour of bird flocks. A swarm of N particles explores a D -dimensional search space, where each particle i maintains a position $\mathbf{X}_i$ and velocity $\mathbf{V}_i$. At each iteration, the velocity is updated based on the particle's personal best position ($\mathbf{P}_i$) and the global best position ($\mathbf{G}$):

$$\mathbf{V}_i(t+1) = w\mathbf{V}_i(t) + c_1r_1(\mathbf{P}_i - \mathbf{X}_i(t)) + c_2r_2(\mathbf{G} - \mathbf{X}_i(t)) \quad (2.4)$$

$$\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \mathbf{V}_i(t+1) \quad (2.5)$$

where w is the inertia weight, c_1 and c_2 are cognitive and social acceleration coefficients, and r_1, r_2 are uniform random numbers in $[0, 1]$.

Standard PSO can suffer from premature convergence when the inertia weight or acceleration coefficients are not carefully tuned. The Phasor PSO variant addresses this limitation by replacing the scalar acceleration parameters with periodic phasor functions.

2.5.2 Phasor PSO Formulation

Ghasemi et al. (Ghasemi et al. 2019) proposed PPSO as a variant of PSO that uses a phase angle θ_i to modulate the velocity update. The key equations are:

Velocity Update:

$$\mathbf{V}_i(t) = |\cos \theta_i(t)|^2 \sin \theta_i(t) \cdot (\mathbf{P}_i(t) - \mathbf{X}_i(t)) + |\sin \theta_i(t)|^2 \cos \theta_i(t) \cdot (\mathbf{G}(t) - \mathbf{X}_i(t)) \quad (2.6)$$

The first term represents the *cognitive component* (attraction toward personal best), and the second term represents the *social component* (attraction toward global best). The phasor terms $|\cos \theta|^2 \sin \theta$ and $|\sin \theta|^2 \cos \theta$ create an oscillatory balance between exploration and exploitation as the phase angle evolves.

Position Update:

$$\mathbf{X}_i(t+1) = \mathbf{X}_i(t) + \mathbf{V}_i(t) \quad (2.7)$$

Phase Angle Update:

$$\theta_i(t+1) = \theta_i(t) + |\cos \theta_i(t) + \sin \theta_i(t)| \times 2\pi \quad (2.8)$$

Velocity Clamping:

$$V_{i,\max}(t+1) = |\cos \theta_i(t)|^2 \times (X_{\max} - X_{\min}) \quad (2.9)$$

The periodic nature of the phasor functions eliminates the need to tune inertia weight and acceleration coefficients manually. When $\cos \theta$ dominates, the cognitive component is emphasised (exploitation); when $\sin \theta$ dominates, the social component drives exploration. This self-adaptive mechanism allows PPSO to balance exploration and exploitation throughout the optimisation process without user intervention (Ghasemi et al. 2019).

2.5.3 Advantages of PPSO over Standard PSO

- **Parameter-free:** PPSO does not require tuning of inertia weight or acceleration coefficients.
- **Oscillatory exploration:** The phasor terms create natural oscillations that help particles escape local optima.
- **Convergence speed:** Zhang et al. (Zhang, Y. Chen, and Yan 2023) showed that PPSO converges faster than FOA, SBO, BRO, MFO, and ALO in CatBoost hyperparameter optimisation.
- **Simplicity:** The algorithm introduces only one additional variable (phase angle θ) per particle.

2.6 The Hybrid CatBoost-PPSO Model

The hybrid CatBoost-PPSO model, first proposed by Zhang et al. (Zhang, Y. Chen, and Yan 2023), combines the predictive power of CatBoost with the automated hyperparameter tuning capability of PPSO. The integration follows a wrapper approach:

1. **Initialise PPSO:** Generate an initial swarm of particles, where each particle's position vector encodes a set of CatBoost hyperparameters (iterations, learning rate, depth, L_2 regularisation).
2. **Evaluate Fitness:** For each particle, train a CatBoost model with the encoded hyperparameters using K -fold time-series cross-validation, and compute the mean validation RMSE as the fitness score.
3. **Update Particles:** Apply the PPSO velocity and position update rules (Equations 2.6–2.9) to move particles toward better hyperparameter configurations.

4. **Iterate:** Repeat steps 2–3 until the maximum number of iterations is reached or early stopping is triggered.
5. **Final Training:** Train the final CatBoost model on the full training set using the globally best hyperparameters found by PPSO.

This approach is advantageous over grid search or random search because PPSO intelligently explores the hyperparameter space, converging toward promising regions while maintaining diversity through its phasor-driven exploration mechanism. The use of time-series cross-validation within the fitness function ensures that no future data leaks into the training process, preserving the temporal integrity of the evaluation.

2.7 Evaluation Metrics

To comprehensively assess model performance, six statistical metrics are used, following the methodology of Zhang et al. (Zhang, Y. Chen, and Yan 2023):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2.10)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2.11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2.12)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (2.13)$$

$$\text{RAE} = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{\sum_{i=1}^n |y_i - \bar{y}|} \quad (2.14)$$

$$\text{WI} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (|\hat{y}_i - \bar{y}| + |y_i - \bar{y}|)^2} \quad (2.15)$$

where y_i is the i -th observed value, $\hat{y}_i$ is the i -th predicted value, $\bar{y}$ is the mean of observations, and n is the number of test samples. MAE, RMSE, MAPE, and RAE are error metrics (lower is better), while R^2 and WI are goodness-of-fit metrics (higher is better, with a maximum of 1.0). The Willmott Index (WI) (Willmott 1982) is included because it is less sensitive to outliers than R^2 and provides a bounded measure of agreement between observed and predicted values.

2.8 Related Work

Table 2.2 summarises key studies in weather-driven electricity demand forecasting that informed the design of this project.

Table 2.2: Summary of related work in electricity demand forecasting.

Study	Method	Data	Key Finding
Al-Musaylh et al. (Al-Musaylh et al. 2018)	MARS, SVR, ARIMA	Aggregated demand, Australia	SVR and MARS outperform ARIMA for short-term forecasting
Khan et al. (Khan, Byun, et al. 2020)	XGBoost + CatBoost + RF	Building energy	Hybrid ensemble improves over individual models
Khan et al. (Khan, Kim, et al. 2021)	CatBoost + XGBoost + MLP	Building energy	Weather, holidays, weekends are key influencing factors
Li et al. (Li et al. 2020)	Cubist + PSO	University buildings	PSO-optimised model improves accuracy by 42.2%
Zhang et al. (Zhang, Y. Chen, and Yan 2023)	CatBoost-PPSO	Hourly residential, 1 customer	PPSO yields best test-set RMSE among 6 optimisers
Nair (Nair 2025)	RF, XGBoost, CNN-LSTM-Attn	Hourly household, Finland	Segmentation-based forecasting improves interpretability

The present study differs from prior work in three important ways: (1) it applies the CatBoost-PPSO methodology to *daily national-level* demand rather than hourly single-customer data; (2) it focuses on a *tropical equatorial* climate where cooling demand dominates; and (3) it uses a broader set of 16 engineered features including a Comfort Climate Index and autoregressive lag/rolling statistics.



Figure 2.1: Daily electricity demand time series for Singapore (February 2023 – December 2025), showing seasonal patterns and day-to-day variability.

Figure 2.1 shows the daily electricity demand time series used in this study. The data exhibits clear weekly cyclicality (lower weekend demand), seasonal variation (higher demand during drier, hotter months), and occasional demand drops associated with public holidays.

2.9 Conclusions

This chapter has reviewed the theoretical foundations of electricity demand forecasting, from classical statistical methods to modern machine learning approaches. The CatBoost algorithm and its native handling of categorical features make it well-suited for the multi-type feature set used in this project. The PPSO meta-heuristic provides an efficient, parameter-free mechanism for optimising CatBoost’s hyperparameters. The

six evaluation metrics adopted from the reference study enable a rigorous and multi-dimensional comparison of model performance. The following chapter presents the system design that operationalises these methods into a cohesive forecasting pipeline.

Chapter 3

Design

This chapter presents the architectural design of the electricity demand forecasting system. It describes the overall pipeline, data acquisition strategy, feature engineering approach, model selection rationale, and the design of the PPSO-based hyperparameter optimisation process.

3.1 System Architecture Overview

The forecasting system is designed as a modular, end-to-end machine learning pipeline comprising six sequential stages: configuration loading, data acquisition, preprocessing and feature engineering, train/test splitting, model training and hyperparameter tuning, and evaluation with visualisation. Figure 3.1 presents the high-level pipeline architecture.

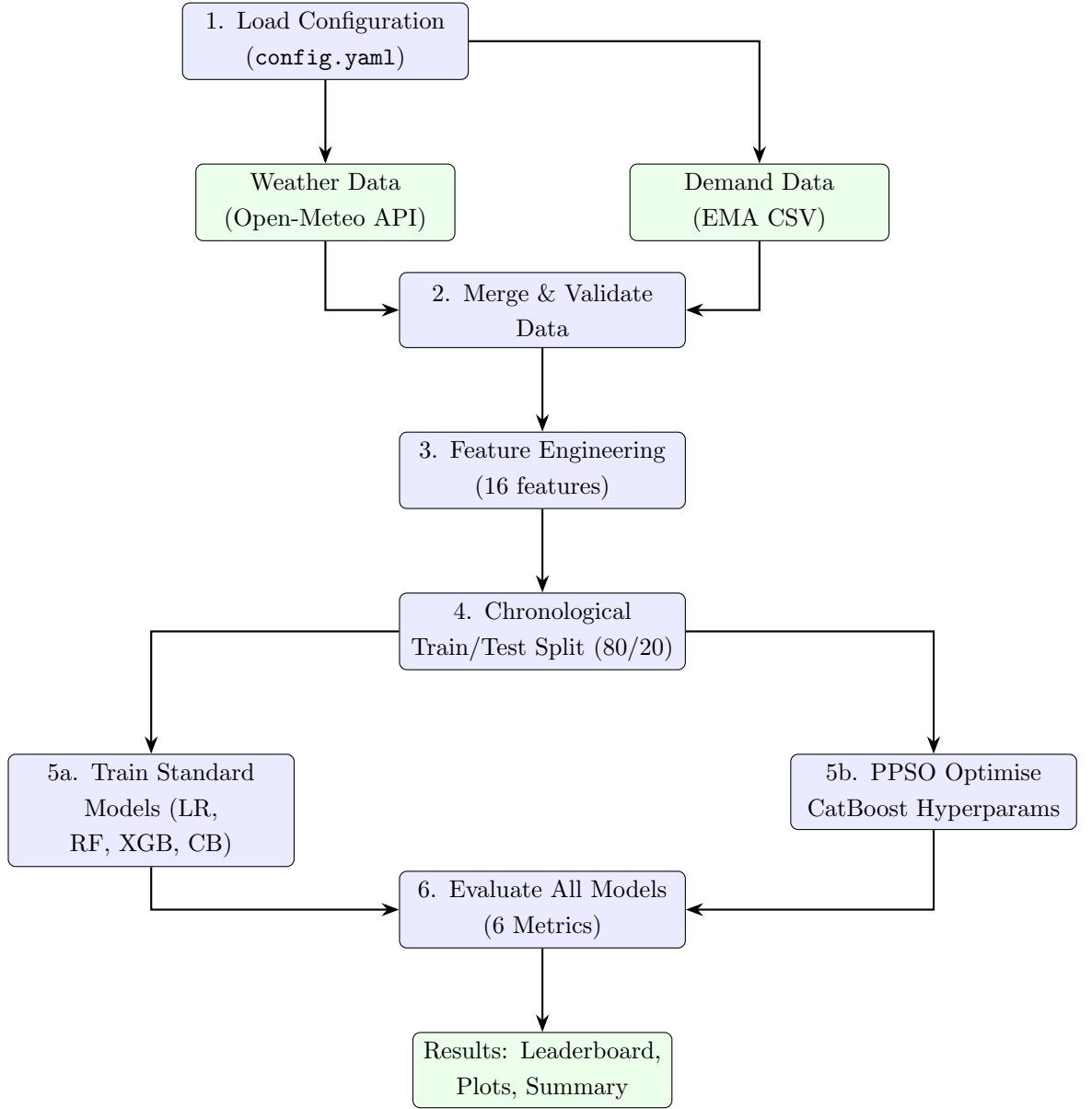


Figure 3.1: High-level architecture of the electricity demand forecasting pipeline.

Each stage is implemented as an independent Python module, enabling modularity, testability, and ease of modification. A single YAML configuration file controls all pipeline parameters, from data paths and weather API coordinates to model hyperparameter bounds and output directories.

3.2 Data Acquisition Design

3.2.1 Electricity Demand Data

Daily electricity demand data for Singapore is obtained from the Energy Market Authority (EMA) (Energy Market Authority of Singapore [2025](#)). The data is stored as a CSV file (`ema_daily_demand.csv`) with two columns: `date` (YYYY-MM-DD) and `demand_mwh` (total daily demand in megawatt-hours). The dataset spans from January 2023 to December 2025, providing 1,096 daily observations.

3.2.2 Weather Data

Historical weather observations are retrieved programmatically from the Open-Meteo Archive API (Open-Meteo [2025](#)), which provides free access to reanalysis and station-interpolated meteorological data. The API is queried with Singapore’s geographic coordinates (1.3521°N, 103.8198°E) to retrieve four daily weather variables:

Table 3.1: Weather variables retrieved from Open-Meteo API.

Variable	API Parameter	Unit
Mean temperature	<code>temperature_2m_mean</code>	°C
Mean relative humidity	<code>relative_humidity_2m_mean</code>	%
Maximum wind speed	<code>wind_speed_10m_max</code>	m/s
Cumulative solar radiation	<code>shortwave_radiation_sum</code>	W/m ²

3.2.3 Data Integration

The weather and demand datasets are merged on the `date` column using an inner join, ensuring that only dates with both weather observations and demand records are retained. After merging, the dataset is sorted chronologically and validated (minimum 31 rows required). The merged dataset contains 1,065 daily observations spanning February 2023 to December 2025.

3.3 Feature Engineering Design

A total of 16 features are constructed from the raw data across four categories. Table [3.2](#) provides a complete description.

Table 3.2: Complete feature set used for demand forecasting (16 features).

Category	Feature	Type	Description
Weather	temperature	Numeric	Daily mean temperature ($^{\circ}\text{C}$)
	humidity	Numeric	Daily mean relative humidity (%)
	wind_speed	Numeric	Daily max wind speed (m/s)
	solar_radiation	Numeric	Cumulative radiation (W/m^2)
Derived	cci	Numeric	Comfort Climate Index (Eq. 2.1)
	season	Categorical	Season (1–4, derived from month)
Temporal	day_of_week	Categorical	0 (Mon) to 6 (Sun)
	month	Categorical	1–12
	is_weekend	Binary	1 if Saturday/Sunday
	is_holiday	Binary	1 if Singapore public holiday
Autoregressive	demand_lag_1	Numeric	Demand 1 day ago
	demand_lag_7	Numeric	Demand 7 days ago
	demand_rolling_mean_3	Numeric	3-day rolling mean demand
	demand_rolling_mean_7	Numeric	7-day rolling mean demand
	demand_rolling_std_3	Numeric	3-day rolling std of demand
	demand_rolling_std_7	Numeric	7-day rolling std of demand

3.3.1 Rationale for Feature Categories

- **Weather features** provide the exogenous drivers of cooling demand. Temperature and humidity are expected to be the strongest weather predictors given Singapore’s air-conditioning-dominated load profile.
- **Derived features:** The CCI captures the combined effect of temperature and humidity on thermal comfort (Equation 2.1). The season variable provides a coarse temporal grouping that may capture monsoon-related demand patterns.
- **Temporal features** encode calendar effects. Weekend and holiday indicators capture the well-documented demand reduction on non-working days, while day-of-week and month allow the model to learn finer-grained periodic patterns.
- **Autoregressive features** incorporate demand history. Lag-1 captures the strong day-to-day autocorrelation of demand, while lag-7 captures weekly periodicity. Rolling means and standard deviations smooth out noise and provide information about recent demand trends and volatility.

The construction of lag and rolling features introduces missing values at the beginning of the time series (up to 7 days for lag-7, up to 7 days for rolling-7). These rows

are dropped after feature engineering, resulting in the final dataset of 1,065 usable observations.

3.4 Model Selection Rationale

Five models were selected to provide a progression from simple to complex:

1. **Linear Regression:** Serves as an interpretable baseline. Features are standardised using `StandardScaler` before fitting.
2. **Random Forest:** Captures non-linear relationships and interactions without requiring feature scaling. Provides feature importance via impurity reduction.
3. **XGBoost:** Sequential boosting with regularisation. Tests whether gradient boosting improves over bagging (RF).
4. **CatBoost (default):** Native categorical feature handling. Tests whether ordered boosting and target statistics improve over XGBoost for this dataset.
5. **CatBoost-PPSO:** CatBoost with PPSO-optimised hyperparameters. Tests the value of automated meta-heuristic tuning.

This progression allows us to isolate the contribution of each methodological improvement: non-linearity (LR $\rightarrow$ RF), sequential boosting (RF $\rightarrow$ XGBoost), categorical handling (XGBoost $\rightarrow$ CatBoost), and hyperparameter optimisation (CatBoost $\rightarrow$ CatBoost-PPSO).

3.5 PPSO Hyperparameter Tuning Design

The PPSO optimiser is configured to search over a four-dimensional hyperparameter space (Table 2.1). The design choices are as follows:

- **Swarm size:** 15 particles. Balances exploration breadth with computational cost.
- **Maximum iterations:** 20 PPSO iterations. Each iteration evaluates all 15 particles.
- **Fitness function:** Mean RMSE from 3-fold `TimeSeriesSplit` cross-validation on the training set. Time-series splitting ensures that each validation fold contains only future data relative to its training fold, preventing temporal leakage.
- **Early stopping:** PPSO terminates if the global best fitness does not improve for 5 consecutive iterations. CatBoost also uses internal early stopping (20 rounds)

within each cross-validation fold.

- **Velocity clamping:** Velocities are clamped using the phasor-modulated maximum velocity (Equation 2.9) to prevent particles from overshooting the search bounds.

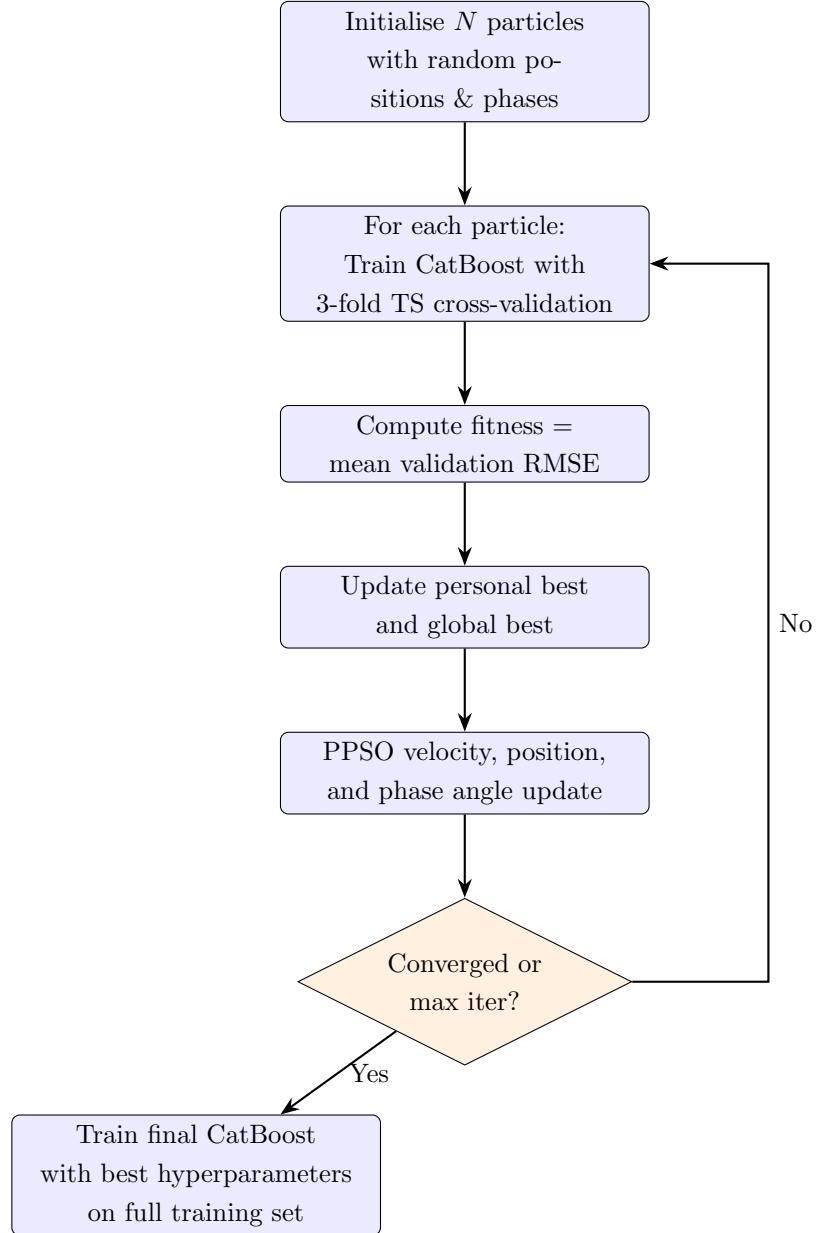


Figure 3.2: PPSO hyperparameter optimisation workflow for CatBoost.

3.6 Train/Test Split Strategy

A chronological 80/20 split is employed: the first 852 observations form the training set, and the remaining 213 observations form the test set. Unlike random splitting, chronological splitting preserves the temporal ordering of the data and prevents future information from leaking into the training process. This is essential for time-series forecasting, where the model must generalise to unseen future dates.

Table 3.3: Train/test split statistics.

Partition	Samples	Proportion	Date Range
Training set	852	80%	Feb 2023 – Jun 2025
Test set	213	20%	Jul 2025 – Dec 2025
Total	1,065	100%	Feb 2023 – Dec 2025

3.7 Exploratory Data Analysis

Before model training, exploratory data analysis (EDA) was conducted to understand the structure and relationships within the dataset.

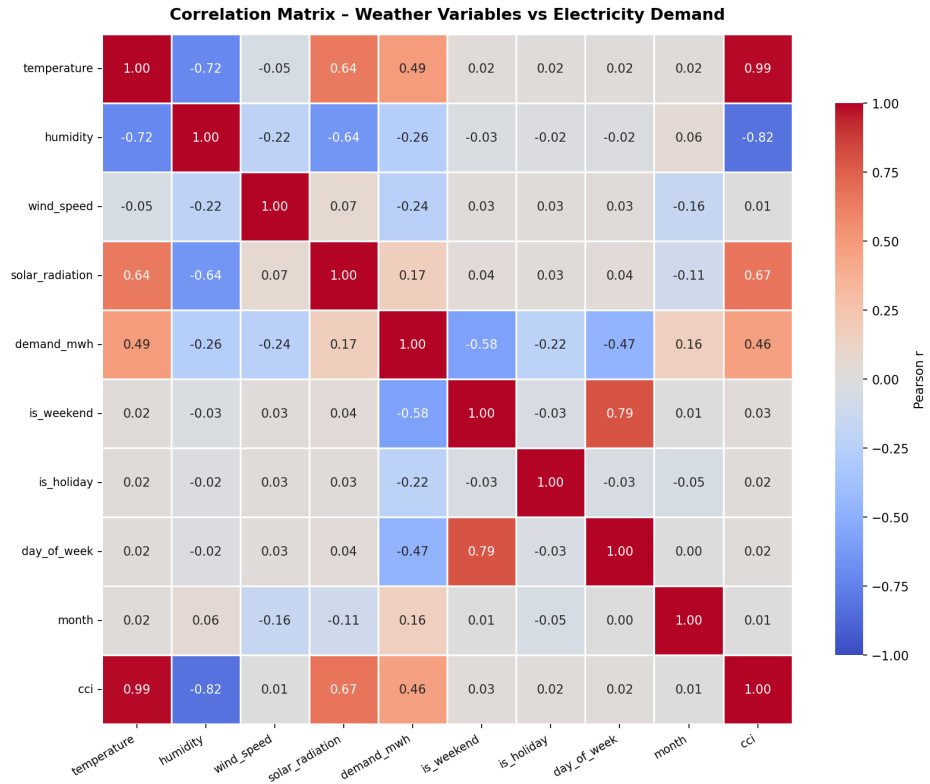


Figure 3.3: Pearson correlation heatmap of all numeric features and the target variable (**demand_mwh**). Lag-1 demand shows the strongest correlation with the target, followed by rolling mean features and temperature.

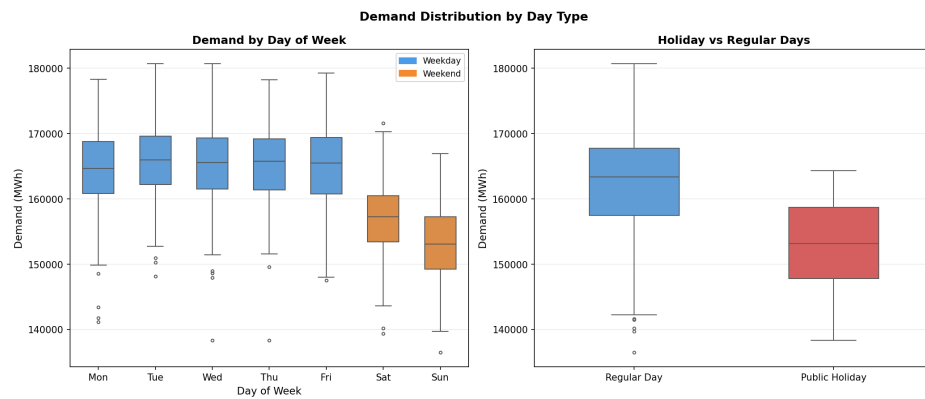


Figure 3.4: Distribution of daily electricity demand by day type (weekday vs. weekend/holiday). Weekday demand is consistently higher, confirming the importance of temporal calendar features.

3.8 Conclusions

This chapter has presented the design of a modular forecasting pipeline that integrates weather and demand data, constructs 16 engineered features across four categories, trains five progressively complex models, and employs PPSO for automated hyperparameter tuning. The chronological train/test split ensures evaluation integrity. The following chapter describes the technical implementation of each component.

Chapter 4

Implementation

This chapter describes the technical implementation of the forecasting pipeline, covering the technology stack, code structure, and the detailed implementation of each module—from data loading through model training to evaluation.

4.1 Technology Stack

The pipeline is implemented entirely in Python 3 using the libraries listed in Table 4.1. All experiments were run on a local machine; no GPU acceleration was required as the models are tree-based.

Table 4.1: Technology stack and key libraries.

Category	Library	Purpose
Data handling	<code>pandas</code> , <code>numpy</code>	DataFrame operations, numeric computation
Weather API	<code>requests</code>	HTTP requests to Open-Meteo
ML framework	<code>scikit-learn</code>	Linear Regression, Random Forest, scaling, CV
Gradient boosting	<code>xgboost</code>	XGBoost regressor
Gradient boosting	<code>catboost</code>	CatBoost regressor
Calendar features	<code>holidays</code>	Singapore public holiday detection
Configuration	<code>pyyaml</code>	YAML configuration parsing
Visualisation	<code>matplotlib</code> , <code>seaborn</code>	Plots and charts
Serialisation	<code>pickle</code>	Model persistence

4.2 Project Structure

The codebase is organised into a modular directory structure:

```
1 electricity-demand/
2   configs/
3     config.yaml      # Pipeline configuration
4   data/
5     ema_daily_demand.csv # Input demand data
6   src/
7     data_loader.py    # Weather API + CSV loading
8     preprocessing.py  # Feature engineering
9     model.py          # Model definitions + PPSO
10    trainer.py         # Training wrapper
11    evaluator.py       # Metrics + plotting
12    visualizer.py      # EDA visualisation
13  main.py              # Pipeline entry point
14  results/             # Generated outputs
15    plots/             # Visualisation images
16    model_comparison.csv # Leaderboard
17    summary.yaml       # Summary statistics
```

Listing 4.1: Project directory structure.

4.3 Data Loading Module

The `data_loader.py` module implements two data-loading functions and a merging function:

4.3.1 Weather Data Retrieval

The `fetch_weather_data()` function queries the Open-Meteo Archive API with the following parameters:

- **Endpoint:** `https://archive-api.open-meteo.com/v1/archive`
- **Coordinates:** Latitude 1.3521, Longitude 103.8198 (Singapore)
- **Timezone:** Asia/Singapore
- **Variables:** `temperature_2m_mean, relative_humidity_2m_mean, wind_speed_10m_max, shortwave_radiation_sum`

The response JSON is parsed into a DataFrame with columns `date`, `temperature`, `humidity`, `wind_speed`, and `solar_radiation`. Any missing values are forward-filled to ensure continuity.

4.3.2 Demand Data Loading

The `load_demand_data()` function reads the EMA CSV file, validates the presence of the `demand_mwh` column, and returns a sorted DataFrame.

4.3.3 Data Merging

The `load_all_data()` function orchestrates both data sources, performs an inner join on the `date` column, and validates that the merged dataset contains at least 31 observations.

4.4 Preprocessing Module

The `DataPreprocessor` class in `preprocessing.py` applies three sequential feature engineering steps:

4.4.1 Weather-Engineered Features

The Comfort Climate Index (CCI) is computed when both temperature and humidity columns are present:

```
1 df['cci'] = T + 0.55 * (1 - (RH / 100.0)) * (T - 14.5)
```

Listing 4.2: CCI computation.

4.4.2 Temporal Features

Calendar-based features are extracted from the `date` column:

```
1 df['day_of_week'] = df['date'].dt.dayofweek # 0=Mon, 6=Sun
2 df['month'] = df['date'].dt.month # 1-12
3 df['is_weekend'] = df['day_of_week'].isin([5, 6]).astype(int)
4 df['is_holiday'] = df['date'].apply(
5     lambda x: int(x in sg_holidays))
6 df['season'] = (df['month'] % 12 // 3 + 1).astype(int)
```

Listing 4.3: Temporal feature extraction.

Public holidays are detected using the `holidays.SG()` class, which provides a comprehensive list of Singapore gazetted holidays for years 2023–2026.

4.4.3 Lag and Rolling Features

Autoregressive features are computed using shifted and rolling-window operations:

```

1 for lag in [1, 7]:
2     df[f'demand_lag_{lag}'] = df['demand_mwh'].shift(lag)
3
4 for window in [3, 7]:
5     shifted = df['demand_mwh'].shift(1)
6     df[f'demand_rolling_mean_{window}'] = (
7         shifted.rolling(window=window).mean())
8     df[f'demand_rolling_std_{window}'] = (
9         shifted.rolling(window=window).std())

```

Listing 4.4: Lag and rolling feature construction.

The `.shift(1)` applied before rolling ensures that only past data is used, preventing any look-ahead bias. Rows with NaN values introduced by lag and rolling operations are dropped at the end of the preprocessing step.

4.5 Model Implementations

All models are instantiated through a factory function `create_model(model_type)`, which returns the appropriate model object.

4.5.1 Linear Regression

Scikit-learn's `LinearRegression` is used with default parameters. Input features are standardised using `StandardScaler` within the `ModelTrainer` wrapper.

4.5.2 Random Forest

`RandomForestRegressor` is configured with `random_state=42` and `n_jobs=-1` for reproducibility and parallel execution. Default hyperparameters are used, as the purpose is to establish a strong non-linear baseline.

4.5.3 XGBoost

`xgb.XGBRegressor` is configured with `random_state=42` and `n_jobs=-1`. Default hyperparameters are used.

4.5.4 CatBoost (Default)

`cb.CatBoostRegressor` is configured with the following categorical feature declarations:

```

1 categorical_cols = ['season', 'day_of_week', 'month',
2                     'is_weekend', 'is_holiday']

```

```

3 model = cb.CatBoostRegressor(
4     random_state=42,
5     cat_features=categorical_cols)

```

Listing 4.5: CatBoost categorical feature configuration.

This allows CatBoost to use its native Target Statistics encoding rather than one-hot encoding, preserving ordinal information and reducing dimensionality.

4.5.5 CatBoost-PPSO

The `CatBoostPPSOModel` class implements the hybrid model. Its core components are:

PPSO Optimiser

The PPSO class implements the optimisation loop described in Section 2.5:

1. **Initialisation:** N particles are uniformly sampled within the hyperparameter bounds. Phase angles θ are initialised uniformly in $[0, 2\pi]$.
2. **Velocity update:** The PPSO velocity equation (Equation 2.6) is applied element-wise:

```

1 cos_t = np.cos(Theta[i])
2 sin_t = np.sin(Theta[i])
3 term1 = (np.abs(cos_t)**2) * sin_t * (pbest_X[i] - X[i])
4 term2 = (np.abs(sin_t)**2) * cos_t * (gbest_X - X[i])
5 V[i] = term1 + term2

```

Listing 4.6: PPSO velocity update implementation.

3. **Velocity clamping:** Maximum velocity is modulated by the cosine phasor term:

```

1 v_max = (np.abs(cos_t)**2) * (bounds[:, 1] - bounds[:, 0])
2 v_max = np.where(v_max < 1e-6,
3     0.1 * (bounds[:, 1] - bounds[:, 0]), v_max)
4 V[i] = np.clip(V[i], -v_max, v_max)

```

Listing 4.7: PPSO velocity clamping.

4. **Position and phase update:**

```

1 X[i] = np.clip(X[i] + V[i], bounds[:, 0], bounds[:, 1])
2 Theta[i] = Theta[i] + np.abs(cos_t + sin_t) * (2 * np.pi)

```

Listing 4.8: PPSO position and phase angle update.

5. **Early stopping:** If the global best fitness does not improve for 5 consecutive iterations, the loop terminates.

Objective Function

The fitness of each particle is evaluated by training a CatBoost model with the particle’s encoded hyperparameters and computing the mean RMSE across 3-fold `TimeSeriesSplit` cross-validation:

```

1 def _objective_function(self, params):
2     n_estimators = int(np.round(params[0]))
3     learning_rate = float(params[1])
4     depth = int(np.round(params[2]))
5     l2_leaf_reg = float(params[3])
6
7     tscv = TimeSeriesSplit(n_splits=3)
8     cv_rmse_scores = []
9     for train_idx, val_idx in tscv.split(self.X_train):
10         model = cb.CatBoostRegressor(
11             iterations=n_estimators, depth=depth,
12             learning_rate=learning_rate,
13             l2_leaf_reg=l2_leaf_reg, ...)
14         model.fit(X_fold_train, y_fold_train,
15                 eval_set=(X_fold_val, y_fold_val),
16                 early_stopping_rounds=20, verbose=0)
17         fold_rmse = np.sqrt(mean_squared_error(
18             y_fold_val, model.predict(X_fold_val)))
19         cv_rmse_scores.append(fold_rmse)
20     return np.mean(cv_rmse_scores)

```

Listing 4.9: PPSO objective function (fitness evaluation).

Each PPSO iteration therefore involves $N \times K = 15 \times 3 = 45$ CatBoost training runs. With early stopping limiting most runs to fewer than the maximum iterations, the total optimisation time remains practical.

4.6 Training Wrapper

The `ModelTrainer` class provides a unified interface for all models:

- **Scaling:** Optional `StandardScaler` is applied for models that benefit from normalised features (Linear Regression). Tree-based models receive unscaled data.
- **Training:** The `fit()` method delegates to the model’s own `fit()` method, passing any additional keyword arguments (e.g. `param_bounds` for CatBoost-PPSO).

- **Prediction:** The `predict()` method applies the same scaling transform (if any) before calling the model's `predict()`.
- **Persistence:** Trained models, scalars, and feature names are serialised to disk using `pickle`.

4.7 Evaluation Module

The `RegressionEvaluator` class computes all six metrics described in Section 2.7. The implementation uses `scikit-learn`'s `mean_absolute_error`, `mean_squared_error`, and `r2_score`, supplemented by custom implementations for MAPE, RAE, and WI:

```

1 epsilon = np.finfo(np.float64).eps
2 mape = np.mean(np.abs((y_true - y_pred) /
3                 (y_true + epsilon))) * 100
4 rae = np.sum(np.abs(y_true - y_pred)) / (
5     np.sum(np.abs(y_true - np.mean(y_true))) + epsilon)
6 wi_num = np.sum((y_pred - y_true)**2)
7 wi_den = np.sum((np.abs(y_pred - np.mean(y_true)) +
8                 np.abs(y_true - np.mean(y_true)))**2)
9 wi = 1 - (wi_num / (wi_den + epsilon))

```

Listing 4.10: Custom metric implementations.

A machine epsilon (`eps`) is added to denominators to prevent division-by-zero errors.

4.8 Pipeline Orchestration

The `main.py` script orchestrates the entire pipeline in seven sequential steps:

1. Load configuration from `configs/config.yaml`.
2. Load and merge weather + demand data.
3. Run preprocessing and feature engineering.
4. Execute EDA visualisation.
5. Perform chronological train/test split.
6. Train all models (including PPSO optimisation for CatBoost-PPSO).
7. Evaluate all models, generate leaderboard, and save results.

The entire pipeline is executed via a single command: `python main.py`.

4.9 Interactive Forecasting Dashboard

Beyond the offline experimentation pipeline, a lightweight web-based portal was developed to make the trained model accessible to non-technical stakeholders. The portal consists of two pages: a *History Report* page for exploring past demand and the model’s retrospective accuracy over a user-selected period, and a *Demand Trends* page that produces forward-looking demand outlooks by combining the trained CatBoost-PPSO model with live weather forecasts from the Open-Meteo API (Open-Meteo [2025](#)).

4.9.1 Architecture

The dashboard follows a conventional two-tier client–server architecture:

- **Backend (FastAPI, Python):** A thin REST service that loads the serialised `catboost_ppso_model.pkl` at start-up and exposes four endpoints: `/api/summary` (model metadata and test-set metrics), `/api/range` (historical demand, per-day model predictions, and user-friendly statistics for a selected period of at least 30 days), `/api/predict` (single-date forecast), and `/api/forecast` (multi-day forward outlooks driven by Open-Meteo weather forecasts).
- **Frontend (React + Vite):** A single-page application with tab-based navigation between the two pages. Charts are drawn with the Recharts library, and the layout uses a card-based grid with a restrained colour palette to keep the interface clean and legible. Technical metric names are translated into plain language for general users (e.g. “forecast accuracy” in place of R^2 /MAPE, and human-readable feature names such as “Yesterday’s demand” in place of `demand_lag_1`).

4.9.2 Weather Forecast Integration (Open-Meteo)

The Demand Trends page requires weather values for future dates, which are obtained from two Open-Meteo services (Open-Meteo [2025](#)):

- **Short/medium range (up to 16 days):** the standard forecast endpoint (`api.open-meteo.com/v1/forecast`) is queried with `forecast_days=16` and `timezone=Asia/Singapore`, retrieving daily maximum/minimum temperature, precipitation, humidity, wind speed, and solar radiation. The user may choose between the default `best_match` model selection and the European `ecmwf_ifs025` model.
- **Long range (monthly and seasonal outlooks, up to 6 months):** the climate endpoint (`climate-api.open-meteo.com/v1/climate`) provides seasonal-scale daily estimates. Where neither service covers a requested day, the monthly

climatology of the historical dataset is used as a fallback.

The portal explicitly communicates the accuracy characteristics of numerical weather prediction to the user: forecasts for the first 1–3 days are highly reliable (approximately 85–95% for temperature, wind, and pressure), days 4–7 remain useful for identifying trends (approximately 70–80%), while days 8–16 are indicative only (approximately 50–60%) and are best interpreted as general tendencies rather than precise daily values. Monthly and seasonal outlooks derived from climate models are presented as smoothed trends rather than day-level predictions.

4.9.3 Recursive Demand Prediction

The trained model requires the full 16-feature vector described in Section 4.4, including autoregressive lag and rolling features. For future dates these are not observable, so the backend performs *recursive* (iterated) forecasting: starting from the last observed day in the dataset, the model predicts one day at a time, and each prediction is appended to the demand buffer used to compute the lag and rolling features of subsequent days. Temporal features are derived deterministically from each date (including the Singapore holiday calendar), and weather features are taken from the Open-Meteo services described above.

It should be noted that this deployment strategy compounds two sources of uncertainty — weather forecast error and accumulated autoregressive error — and therefore yields lower accuracy than the retrospective evaluation reported in Chapter 5, degrading progressively with the forecast horizon. This trade-off is acceptable for the portal’s purpose of communicating expected demand trends; the research contribution of this project remains the retrospective comparative study of how weather variables influence electricity demand, for which the evaluation methodology in Chapter 5 is unaffected.

4.9.4 Dashboard Interface

The *History Report* page is organised into four areas: (i) a period selector (date range, month, or year, validated to cover at least 30 days), (ii) summary cards showing average daily use, busiest and quietest days, and forecast accuracy for the selected period, (iii) a time-series chart of actual versus model-predicted demand, and (iv) supporting charts showing average demand by day of week and the model’s most influential factors.

The *Demand Trends* page offers three outlook modes — the next 7 or 16 days, a calendar month, or a season (quarter), limited to 6 months ahead — together with a weather-model selector, a confidence banner that states the expected reliability of the current horizon, the projected demand series (smoothed for long-range outlooks), and

the underlying temperature forecast.

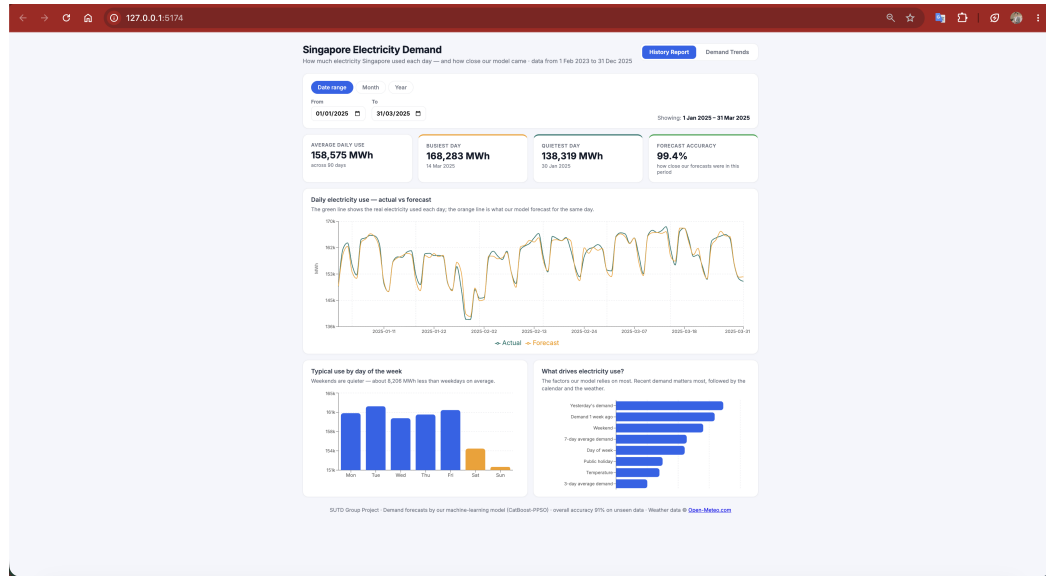


Figure 4.1: History Report page of the dashboard: period selector, summary cards, actual versus predicted demand, and supporting charts for the selected period.

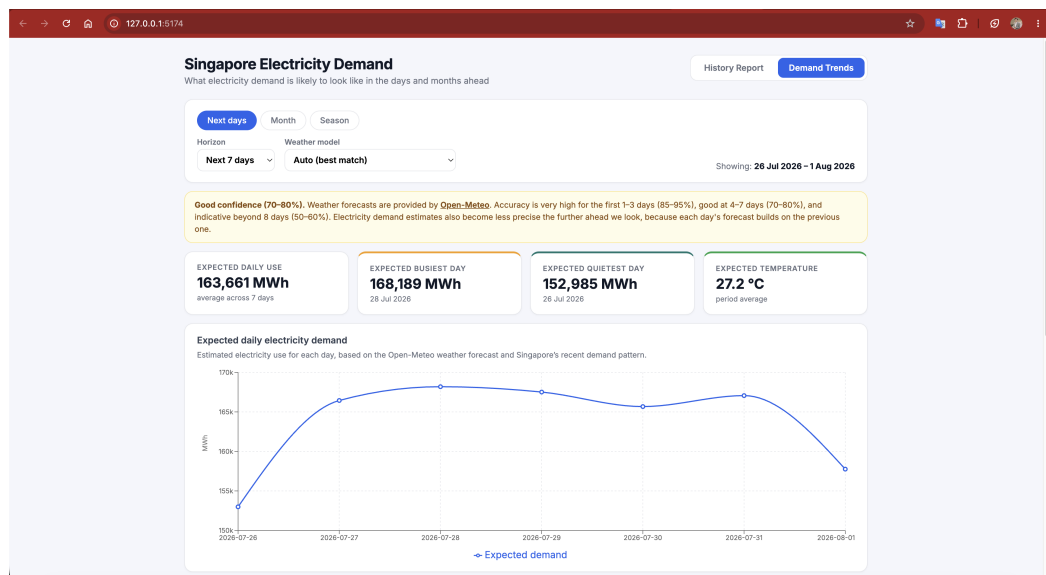


Figure 4.2: Demand Trends page of the dashboard: forward-looking demand outlook driven by Open-Meteo weather forecasts, with an explicit confidence banner reflecting the forecast horizon.

4.10 Conclusions

This chapter has described the implementation of each module in the forecasting pipeline. The code follows a modular design that separates concerns (data loading, preprocessing, modelling, evaluation) and uses a central configuration file for reproducibility. The PPSO implementation closely follows the mathematical formulation of Ghasemi et al. (Ghasemi et al. 2019), with adaptations for CatBoost hyperparameter tuning and time-series cross-validation. In addition, an interactive web dashboard was implemented to expose the trained CatBoost-PPSO model through a simple date-based prediction interface. The following chapter presents the experimental results and comparative evaluation of all five models.

Chapter 5

Evaluation & Testing

This chapter presents the experimental results, provides a comparative analysis of all five forecasting models, examines feature importance, and discusses the findings in the context of the research objectives.

5.1 Experimental Setup

All models were trained on the same 852-sample training set and evaluated on the 213-sample held-out test set using the six metrics defined in Section 2.7. The CatBoost-PPSO model was optimised with 15 particles over 20 iterations, with 3-fold `TimeSeriesSplit` cross-validation as the fitness function. All random seeds were fixed to 42 for reproducibility.

5.2 Model Comparison Results

Table 5.1 presents the performance of all five models on the test set, sorted by RMSE (ascending).

Table 5.1: Model comparison on the test set (213 samples). Models are sorted by RMSE. Best values in each column are shown in **bold**.

Model	MAE	RMSE	R^2	MAPE (%)	RAE	WI
CatBoost-PPSO	1,520.2	1,922.1	0.9111	0.90	0.284	0.9766
Linear Reg.	1,694.0	2,118.9	0.8919	1.01	0.317	0.9731
CatBoost	1,695.1	2,161.0	0.8876	1.00	0.317	0.9700
XGBoost	1,798.6	2,202.0	0.8833	1.07	0.336	0.9691
Random Forest	1,749.3	2,204.5	0.8830	1.04	0.327	0.9665

5.2.1 Key Observations

1. **CatBoost-PPSO achieves the best performance across all six metrics.** Its R^2 of 0.911 indicates that the model explains over 91% of the variance in daily electricity demand. The RMSE of 1,922.1 MWh represents approximately 1.1% of the mean daily demand, and the MAPE of 0.90% confirms excellent relative accuracy.
2. **PPSO optimisation provides a meaningful improvement over default CatBoost.** Comparing CatBoost-PPSO with default CatBoost: MAE improves by 10.3% (1,695 $\rightarrow$ 1,520), RMSE improves by 11.1% (2,161 $\rightarrow$ 1,922), and R^2 improves from 0.888 to 0.911. This demonstrates the value of automated hyperparameter tuning.
3. **Linear Regression performs surprisingly well**, ranking second by RMSE and R^2 . This suggests that the relationship between the engineered features and demand is substantially linear, likely driven by the strong autoregressive features (lag-1 and rolling means). However, Linear Regression cannot capture interaction effects between weather and temporal variables.
4. **XGBoost and Random Forest perform comparably**, with XGBoost showing a slight advantage in MAE and Random Forest in RAE. Both are outperformed by CatBoost-PPSO, suggesting that the native categorical handling and PPSO-tuned hyperparameters of the hybrid model provide genuine advantages.
5. **All models achieve $R^2 > 0.88$ and $WI > 0.96$** , indicating that the 16-feature set captures the dominant drivers of electricity demand effectively.

Model Performance Leaderboard – Singapore Electricity Demand Forecast

Rank	Model	MAE	RMSE	R^2	MAPE	RAE	WI
1	CatBoost-PPSO	1520.2	1922.1	0.9111	0.902	0.2841	0.9766
2	Linear Regression	1694.0	2118.9	0.8919	1.0099	0.3166	0.9731
3	CatBoost	1695.1	2161.0	0.8876	1.0023	0.3168	0.97
4	XGBoost	1798.6	2202.0	0.8833	1.0667	0.3362	0.9691
5	Random Forest	1749.3	2204.5	0.883	1.035	0.327	0.9665

Sorted by RMSE (lower is better) · R^2 , WI: higher is better · MAPE, RAE: lower is better

Figure 5.1: Visual comparison of model performance metrics. CatBoost-PPSO (top row) achieves the best scores across all metrics.

5.3 Prediction Analysis

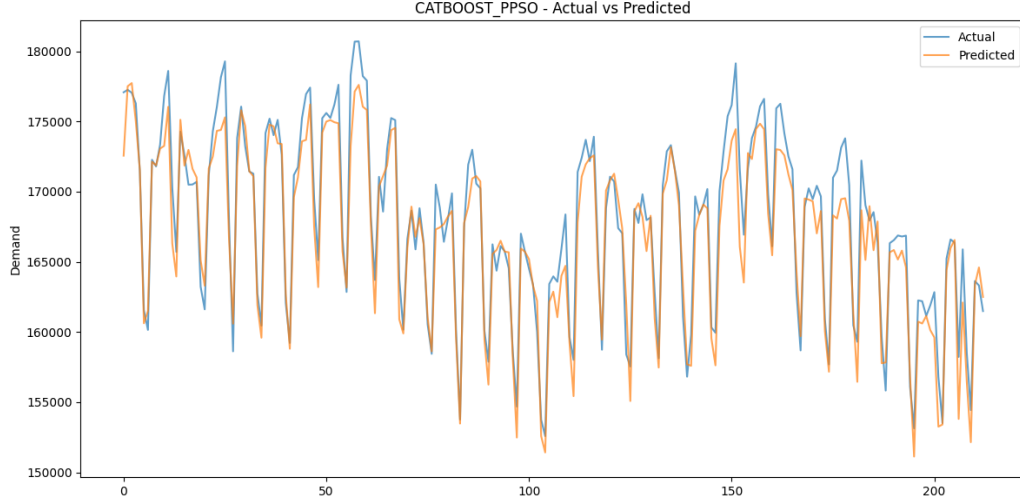


Figure 5.2: Actual vs. predicted daily electricity demand (MWh) for the CatBoost-PPSO model on the test set (213 days). The model closely tracks both the overall demand trend and short-term fluctuations.

The prediction plot reveals that CatBoost-PPSO captures:

- **Weekly cyclicality:** The regular dips on weekends are accurately reproduced.
- **Demand level shifts:** Transitions between higher and lower demand periods are tracked.
- **Occasional outliers:** Some extreme demand drops (likely associated with public holidays) show slightly larger prediction errors, but the overall tracking is tight.

5.4 Feature Importance Analysis

Feature importance was extracted from the CatBoost-PPSO model's underlying CatBoost estimator using built-in feature importance scores.

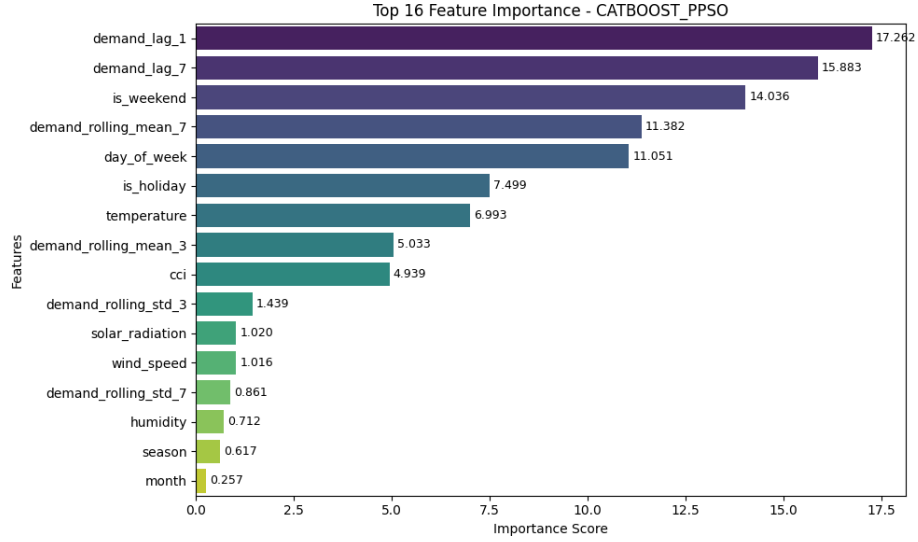


Figure 5.3: Feature importance scores from the CatBoost-PPSO model. The top features are autoregressive demand features, followed by weather and temporal variables.

5.4.1 Interpretation of Feature Importance

The feature importance ranking reveals a clear hierarchy:

1. **Autoregressive features dominate:** `demand_lag_1` (yesterday's demand) is by far the most important predictor, consistent with the strong day-to-day autocorrelation of electricity demand. Rolling means (`demand_rolling_mean_3`, `demand_rolling_mean_7`) and `demand_lag_7` also rank highly, confirming that recent demand history is the strongest predictor of future demand.
2. **Temporal features are significant:** `day_of_week` ranks prominently, reflecting the well-documented weekday/weekend demand differential. `month` and `season` capture longer-term temporal patterns.
3. **Weather features contribute meaningfully:** `temperature` and `humidity` appear in the middle ranks, confirming their role as exogenous demand drivers. The `cci` (Comfort Climate Index) provides additional predictive value by capturing the combined thermal stress effect.
4. **Wind speed and solar radiation** have lower but non-negligible importance, consistent with their indirect influence on cooling demand.

5.4.2 Comparison Across Models

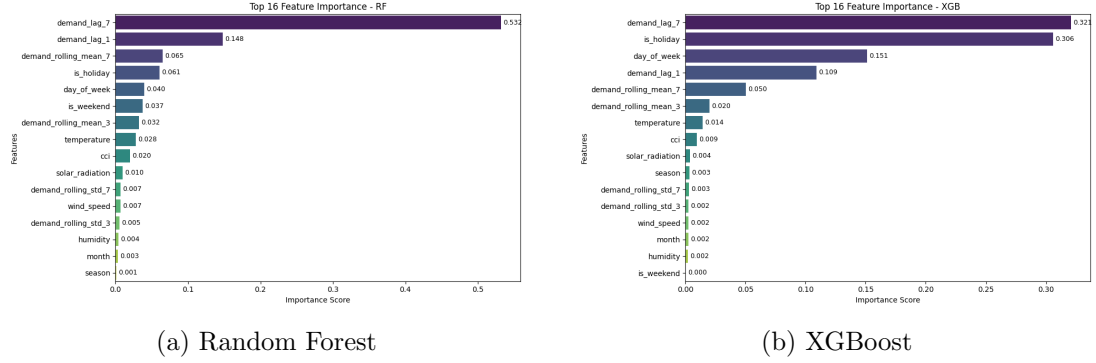


Figure 5.4: Feature importance comparison between Random Forest and XGBoost. Both models agree on the dominance of autoregressive features.

The consistency of feature importance rankings across all tree-based models provides confidence that the identified feature hierarchy is robust and not an artefact of any single algorithm.

5.5 Impact of PPSO Optimisation

To quantify the impact of PPSO hyperparameter tuning, Table 5.2 compares the default CatBoost model with the PPSO-optimised variant.

Table 5.2: Impact of PPSO hyperparameter optimisation on CatBoost performance.

Metric	CatBoost (default)	CatBoost-PPSO	Improvement
MAE (MWh)	1,695.1	1,520.2	−10.3%
RMSE (MWh)	2,161.0	1,922.1	−11.1%
R^2	0.8876	0.9111	+2.6 pp
MAPE (%)	1.00	0.90	−10.0%
RAE	0.317	0.284	−10.4%
WI	0.970	0.977	+0.7 pp

The PPSO-optimised model achieves approximately 10% reduction in all error metrics and a 2.6 percentage-point improvement in R^2 . This confirms the finding of Zhang et al. (Zhang, Y. Chen, and Yan 2023) that PPSO is an effective meta-heuristic for CatBoost hyperparameter optimisation, and demonstrates that the approach generalises from hourly single-customer data to daily national-level demand in a different climate context.

5.6 Comparison with Reference Study

It is informative to compare the results of this project with those of the reference study by Zhang et al. (Zhang, Y. Chen, and Yan 2023). Table 5.3 presents a high-level comparison.

Table 5.3: Comparison between this project and the reference study.

Aspect	Zhang et al. (2023)	This Project
Temporal resolution	Hourly	Daily
Geographic context	Single residential customer	National (Singapore)
Climate	Not specified	Tropical equatorial
Data period	Dec 2010 – Nov 2018	Feb 2023 – Dec 2025
Best model RMSE	42.3 (kW)	1,922.1 (MWh)
Best model R^2	0.971	0.911
Best model WI	0.993	0.977
No. of comparison models	6 hybrid models	5 models (incl. non-hybrid)

The lower R^2 in this project (0.911 vs. 0.971) is expected because daily national-level demand aggregates heterogeneous consumption patterns across millions of consumers, introducing more unexplained variance than a single household’s consumption. Nevertheless, the results confirm that the CatBoost-PPSO methodology is effective across different scales and contexts.

5.7 Discussion

5.7.1 Strengths of the Approach

- The automated PPSO tuning eliminates the need for manual hyperparameter selection, reducing human bias and the risk of sub-optimal configurations.
- The use of time-series cross-validation within the PPSO fitness function preserves temporal integrity and prevents overfitting to the training data.
- The 16-feature set captures multiple dimensions of demand variation (weather, calendar, historical), resulting in consistently high $R^2 > 0.88$ across all models.
- CatBoost’s native categorical handling is advantageous for the multi-type feature set used in this project.

5.7.2 Limitations and Potential Sources of Error

- **Data resolution:** Daily aggregation masks intra-day demand peaks, which are critical for operational grid management.
- **Weather data quality:** Open-Meteo provides reanalysis data that may differ from actual ground-station measurements.
- **External events:** The model does not account for economic shocks, major infrastructure changes, or COVID-19 residual effects on demand patterns.
- **PPSO stochasticity:** Despite fixing the random seed, the stochastic nature of PPSO means that different runs may converge to slightly different hyperparameter configurations.

5.8 Conclusions

The evaluation demonstrates that the CatBoost-PPSO model achieves the best predictive performance among all five tested models, with an R^2 of 0.911, MAE of 1,520.2 MWh, and MAPE of 0.90%. PPSO optimisation provides a consistent $\sim 10\%$ improvement over default CatBoost across all error metrics. Feature importance analysis confirms that autoregressive demand features are the strongest predictors, with temperature and temporal calendar features providing important supplementary signal. These results validate the applicability of the hybrid CatBoost-PPSO methodology to daily national-level demand forecasting in a tropical climate.

Chapter 6

Conclusion

This chapter summarises the main findings of the project, identifies key contributions, acknowledges limitations, and proposes directions for future research.

6.1 Summary of Findings

This project developed and evaluated a comprehensive machine learning pipeline for daily electricity demand forecasting in Singapore, leveraging weather variables and temporal features. Five models were compared—Linear Regression, Random Forest, XGBoost, CatBoost, and CatBoost with Phasor Particle Swarm Optimisation (PPSO)—using data spanning February 2023 to December 2025.

The principal findings are:

1. **CatBoost-PPSO is the best-performing model.** The PPSO-optimised CatBoost model achieved the highest predictive accuracy across all six evaluation metrics, with an R^2 of 0.911, MAE of 1,520.2 MWh, RMSE of 1,922.1 MWh, and MAPE of 0.90% on the held-out test set.
2. **PPSO hyperparameter optimisation provides substantial gains.** Compared to default CatBoost, the PPSO-tuned variant reduced MAE by 10.3%, RMSE by 11.1%, and improved R^2 by 2.6 percentage points. This confirms that automated meta-heuristic tuning yields meaningful improvements over default configurations.
3. **Autoregressive features are the strongest predictors.** Feature importance analysis consistently identified lag-1 demand and rolling-mean features as the

most influential predictors, followed by day-of-week and weather variables (temperature, humidity).

4. **Weather variables contribute meaningfully but are secondary.** Temperature and the derived Comfort Climate Index (CCI) provide useful exogenous signal, particularly for capturing demand increases during hotter periods. However, their importance is subordinate to historical demand patterns at the daily resolution.
5. **The methodology generalises across scales.** The CatBoost-PPSO approach, originally validated on hourly single-customer data by Zhang et al. (Zhang, Y. Chen, and Yan 2023), produces strong results when applied to daily national-level demand in a tropical equatorial climate, demonstrating its versatility.

6.2 Key Contributions

This project makes the following contributions:

- **Application to a new context:** First application of the hybrid CatBoost-PPSO methodology to daily national-level electricity demand in Singapore, demonstrating cross-context generalisability.
- **Comprehensive feature engineering:** Design of a 16-feature set spanning weather, derived comfort indices, temporal calendar effects, and autoregressive demand dynamics, tailored to Singapore’s tropical climate.
- **Rigorous comparative evaluation:** Systematic comparison of five models using six complementary metrics, with time-series cross-validation to prevent temporal leakage.
- **Reproducible pipeline:** An end-to-end Python implementation driven by a single configuration file, enabling easy replication and extension.
- **Interactive forecasting portal:** A React-based web dashboard backed by a FastAPI service, providing a retrospective History Report and a forward-looking Demand Trends outlook that combines the trained CatBoost-PPSO model with live Open-Meteo weather forecasts (Open-Meteo 2025), making the results accessible to non-technical stakeholders.

6.3 Limitations

Several limitations should be noted:

1. **Daily resolution:** The pipeline forecasts total daily demand and does not address intra-day peak load prediction, which is more relevant for operational grid management.
2. **Retrospective evaluation:** Weather data is obtained from the Open-Meteo historical archive rather than a live forecasting service. In a real deployment scenario, the accuracy of weather forecasts would introduce additional prediction uncertainty.
3. **No demand decomposition:** The model predicts aggregate national demand without decomposing it into sectoral (residential, commercial, industrial) contributions. Sector-specific models could provide more targeted insights.
4. **Limited external factors:** The model does not incorporate electricity pricing, economic indicators, or large-scale event calendars (e.g. major sporting events, national emergencies) that may affect demand.
5. **PPSO computational cost:** Although more efficient than grid search, each PPSO iteration requires multiple CatBoost training runs. For very large datasets or high-dimensional hyperparameter spaces, the computational cost may become significant.

6.4 Future Work

Several directions could extend and improve upon this work:

1. **Hourly forecasting:** Extending the pipeline to hourly or sub-hourly resolution would capture intra-day demand patterns and peak load events, which are critical for real-time grid operations.
2. **Deep learning comparison:** Incorporating LSTM, GRU, or Transformer-based models would enable a more comprehensive comparison between tree-based and sequence-based approaches, as suggested by Nair (Nair 2025).
3. **Multi-step forecasting:** Predicting demand for multiple days ahead (e.g. 7-day horizon) would provide greater lead time for generation scheduling and fuel procurement.
4. **Real-time deployment:** Integrating the pipeline with live weather forecast APIs and automated daily retraining would enable operational deployment for day-ahead demand forecasting.
5. **Explainability:** Applying SHAP (SHapley Additive exPlanations) analysis

would provide local, instance-level interpretability in addition to the global feature importance analysis presented in this study.

6. **Transfer learning:** Testing the pipeline on demand data from other tropical cities (e.g. Bangkok, Kuala Lumpur, Jakarta) would evaluate the geographic transferability of the model and feature set.
7. **Hybrid ensemble:** Combining CatBoost-PPSO with other well-performing models (e.g. Linear Regression for its complementary strengths) through stacking or weighted averaging could further improve accuracy.

6.5 Concluding Remarks

This project has demonstrated that the hybrid CatBoost-PPSO model is an effective approach for weather-driven short-term electricity demand forecasting in Singapore. The combination of CatBoost’s native categorical feature handling with PPSO’s parameter-free hyperparameter optimisation produces a forecasting model that is both accurate ($R^2 = 0.911$, MAPE = 0.90%) and practical to implement. The findings contribute to the growing body of evidence that meta-heuristic-optimised machine learning models can outperform standard configurations in energy forecasting applications, and provide a foundation for future work on higher-resolution, real-time demand prediction systems for tropical urban environments.

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